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Targeting the Chemokine System in Rheumatoid Arthritis and Vasculitis

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Abstract

Arrest of circulating leukocytes and subsequent diapedesis is a fundamental component of inflammation. In general, the leukocyte migration cascade is tightly regulated by chemoattractants, such as chemokines. Chemokines, small secreted chemotactic cytokines, as well as their G-protein-coupled seven transmembrane spanning receptors, control the migratory patterns, positioning and cellular interactions of immune cells. Increased levels of chemokines and their receptors are found in the blood and within inflamed tissue in patients with rheumatoid arthritis (RA) and vasculitis. Chemokine ligand-receptor interactions regulate the recruitment of leukocytes into tissue, thus contributing in important ways to the pathogenesis of RA and vasculitis. Despite the fact that blockade of chemokines and chemokine receptors in animal models have yielded promising results, human clinical trials in RA using inhibitors of chemokines and their receptors have generally failed to show clinical benefits. However, recent early phase clinical trials suggest that strategies blocking specific chemokines may have clinical benefits in RA, demonstrating that the chemokine system remains a promising therapeutic target for rheumatic diseases, such as RA and vasculitis and requires further study.

Keywords: Chemokine, Rheumatoid arthritis, Vasculitis

1. Introduction

Egress of leukocytes from the circulation into the extravascular space is a fundamental component of inflammation. Leukocyte recruitment from blood into tissues follows a well-established paradigm: (i) tethering and rolling on the vessel wall, (ii) firm arrest on the endothelium, (iii) spreading out and crawling in all directions on the vessel, and transendothelial migration to extravasate into tissue. This process is tightly controlled by adhesion molecules and chemoattractants including chemokines [\(1\)](#).

Chemokines are chemotactic cytokines that regulate the migration of a wide variety of immune cells [\(2\)](#). In the context of the immune system, chemokines control the positioning of all immune cells and contribute to immune cell development, homeostasis and inflammation. More than 50 chemokines and 19 chemokine receptors have been identified in humans and mice ([Table 1](#)) [\(2\)](#). These small (8-10 kDa) secreted proteins are classified into four groups on the location of cysteine residues near the N terminus of their primary amino acid sequence: XC-chemokines contain a single N-terminal cysteine, CC-chemokines have two adjacent residues, CXC-chemokines have two cysteines separated by one other amino acid, and CX₃C-chemokines have three amino acids between the cysteine residues [\(2\)](#). Most chemokines are secreted into the extracellular space where they bind heparin-like glycosaminoglycans on the cell surface and are embedded in the extracellular matrix, forming transient or stable gradients. Chemokines and their gradients are detected by binding to specific chemokine receptors. Chemokine receptors are seven-transmembrane receptors and are expressed on the surface of immune cells. These receptors are G-protein coupled receptors that signal via Gi-type G proteins. There are approximately 20 *classical* chemokine receptors coupled to G proteins and regulate cell migration, whereas 5 *atypical* chemokine receptors that do not couple to G proteins and do not induce cell migration have been identified ([Table 2](#)) [\(2\)](#).

Table 1.

Chemokines and Their Receptors.

Chemokine	Other names	Receptor
CXCL1	GRO α , MGSA, mouse KC	CXCR2, ACKR1
CXCL2	GRO β , MIP-2 α , mouse MIP2	CXCR2, ACKR1
CXCL3	GRO γ , MIP-2 β	CXCR2, ACKR1
CXCL4	PF4	Unknown
CXCL4L1	PF4V1	Unknown
CXCL5	ENA-78, mouse LIX ^a	CXCR2, ACKR1
CXCL6	GCP-2 (human only)	CXCR1, CXCR2, ACKR1
CXCL7	NAP-2	CXCR2, ACKR1
CXCL8	IL-8 (human only)	CXCR1, CXCR2, ACKR1
CXCL9	Mig	CXCR3
CXCL10	IP-10	CXCR3
CXCL11	I-TAC	CXCR3, ACKR1, ACKR3
CXCL12	SDF-1	CXCR4, ACKR3
CXCL13	BLC, BCA-1	CXCR5, ACKR1, ACKR4
CXCL14	BRAK	Unknown
CXCL15	Lungkine (mouse only)	Unknown
CXCL16		CXCR6
CCL1	I-309, mouse TCA3	CCR8
CCL2	MCP-1, mouse JE	CCR2, ACKR1, ACKR2
CCL3^b	MIP-1 α , LD78 α	CCR1, CCR5, ACKR2
CCL3L1	LD78 β	CCR1, CCR3, CCR5, ACKR2
CCL4	MIP-1 β	CCR5, ACKR2
CCL4L1	LAG-1	CCR5
CCL5	RANTES	CCR1, CCR3, CCR5, ACKR2
CCL6	C-10, MRP-1 (mouse only)	Unknown
CCL7	MCP-3, mouse Fic or MARC	CCR2, CCR3, ACKR1, ACKR2

Chemokine	Other names	Receptor
CCL8	MCP-2	Human: CCR1, CCR2, CCR3, CCR5, ACKR1, ACKR2; mouse: CCR8, ACKR1, ACKR2
CCL9/10	MIP-1 γ , MRP-2 (mouse only)	Unknown
CCL11	Eotaxin-1	CCR3, ACKR2
CCL12	MCP-5 (mouse only)	CCR2
CCL13	MCP-4 (human only)	CCR2, CCR3, CCR5, ACKR1, ACKR2
CCL14	HCC-1 (human only)	CCR1, ACKR1, ACKR2
CCL15	Leukotactin-1, HCC-2, MIP-5 (human only)	CCR1, CCR3
CCL16	HCC-4, NCC-4, LEC (human only)	CCR1, CCR2, CCR5, ACKR1
CCL17	TARC	CCR4, ACKR1, ACKR2
CCL18	PARC, DC-CK1 (human only)	CCR8
CCL19	MIP-3 β , ELC	CCR7, ACKR4
CCL20	MIP-3 α , LARC	CCR6
CCL21	SLC, 6CKine	CCR6, CCR7, ACKR4
CCL22	MDC	CCR4, ACKR1, ACKR2
CCL23	MPIF-1, MMP-3 (human only)	Unknown
CCL24	Eotaxin-2, MPIF-2	CCR3
CCL25	TECK	CCR9, ACKR4
CCL26	Eotaxin-3	CCR3, CX ₃ CR1
CCL27	CTAK	CCR10
CCL28	MEC	CCR3, CCR10
XCL1	Lymphotactin α , SCM-1 α	XCR1
XCL2	Lymphotactin β , SCM-1 β	XCR1
CX₃CL1	Fractalkine	CX ₃ CR1

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Table 2.

Chemokine Receptors: Cellular Expression and Primary Function.

Receptor	Cell expression	General immune function
Classical chemokine receptors		
CXCR1	Neutrophils and lower expression on monocytes, NK cells, mast cells, basophils, CD8 ⁺ T _{EFF} cells and endothelial cells	Neutrophil trafficking
CXCR2	Neutrophils and lower expression on monocytes, NK cells, mast cells, basophils, CD8 ⁺ T cells and endothelial cells	B cell lymphopoiesis, neutrophil egress from bone marrow and neutrophil trafficking
CXCR3	T _H 1 cells, CD8 ⁺ T _{CM} cells and T _{EM} cells, NK cells, NKT cells, plasmacytoid dendritic cells, B cells, T _{reg} cells and T _{FH} cells	Type 1 adaptive immunity
CXCR4	Most leukocytes and endothelial cells	Hematopoiesis, organogenesis and bone marrow homing
CXCR5	B cells, T _{FH} cells, T _{FR} cells CD8 ⁺ T _{EM} cells	B cell and T cell trafficking in lymphoid tissue to the B cell zone or follicles
CXCR6	T _H 1 cells, T _H 17 cells, γδT cells, ILCs, NKT cells, NK cells, plasma cells, endothelial cells	ILC function and adaptive immunity
CCR1	Monocytes, macrophages, neutrophils, T _H 1 cells, basophils, dendritic cells	Innate immunity and adaptive immunity
CCR2	Monocytes, macrophages, T _H 1 cells, immature dendritic cells, basophils, NK cells and endothelial cells	Monocyte trafficking and type 1 adaptive immunity
CCR3	Eosinophils and lower expression on basophils and mast cells	Type 2 adaptive immunity, eosinophil distribution and

Receptor	Cell expression	General immune function
		trafficking
CCR4	T _H 2 cells, skin-homing T cells and lung-homing T cells, T _{reg} cells and lower expression on T _H 17 cells, CD8 ⁺ T cells, monocytes, B cells and immature dendritic cells	Homing of T cells to the skin and lungs and type 2 immune responses
CCR5	Monocytes, macrophages, T _H 1 cells, NK cells, T _{reg} cells, CD8 ⁺ T cells, dendritic cells and neutrophils	Type 1 adaptive immunity
CCR6	T _H 17 cells, immature dendritic cells, $\gamma\delta$ T cells, NKT cells, NK cells, T _{reg} cells and T _{FH} cells	Immature dendritic cell trafficking, GALT development and type 17 adaptive immune responses
CCR7	Naïve T cells, T _{CM} cells, T _{RCM} cells, mature dendritic, B cells	Mature dendritic cell, B cell and T cell trafficking in lymphoid tissue to T cell zone and egress of dendritic cells and T cells from tissue
CCR8	T _H 2 cells, T _{reg} cells, skin T _{RM} cells, $\gamma\delta$ T cells, monocytes and macrophage	Immune surveillance in skin, type 2 adaptive immunity and thymopoiesis
CCR9	Gut-homing T cells, thymocytes, B cells, dendritic cells and plasmacytoid dendritic cells	Homing of T cells to the gut, GALT development and function and thymopoiesis
CCR10	Skin-homing T cells and IgA-plasmablasts	Homing immunity at mucosal sites and immune surveillance in the skin
XCR1	Cross-presenting CD8 ⁺ dendritic cells and thymic dendritic cells	Antigen cross-presentation by CD8 ⁺ dendritic cells
CX ₃ CR1	Resident monocytes, macrophages, microglia, T _H 1 cells, CD8 ⁺ T _{EM} cells, NK cells, $\gamma\delta$ T cells and dendritic cells	Patrolling monocytes in innate immunity, microglial cell and NK cell migration and type 1 adaptive immunity
Atypical chemokine receptors		
ACKR1 (DARC)	Red blood cells and endothelial cells	Chemokine transcytosis and chemokine scavenging

Receptor	Cell expression	General immune function
ACKR2 (D6)	Dendritic cells, B cells and lymphatic endothelium	Chemokine scavenging
ACKR3 (CXCR7)	B cell and stromal cells	Shaping chemokine gradients for CXCR4
ACKR4 (CCRL1; CCX-CKR)	Thymic epithelium	Chemokine scavenging

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DARC, Duffy Antigen Receptor for Chemokines; GALT, gut-associated lymphoid tissue; ILCs, innate lymphoid cells; NK, natural killer; NKT, natural killer T; T_{CM} cell, central memory T cell; T_{EFF} cell, effector T cell; T_{EM} cell, effector-memory T cell; T_{FH} cell, follicular helper T cell; T_{FR} cell, follicular regulatory T cell; T_H cell, T helper; T_{RCM}, recirculating memory T cell; T_{REG} cell, regulatory T cell; T_{RM} cell, resident-memory T cell.

The chemokine system and other inflammatory mediators might play a central role in chronic inflammation such as autoimmune diseases. In fact, high levels of several chemokines are seen in the blood and disease tissues, such as the joints and blood vessels, in patients with rheumatoid arthritis (RA) [\(3\)](#) and vasculitis [\(4\)](#), and their levels have been associated with the disease severity and/or activity. Chemokines and their receptors likely contribute to the recruitment of immune cells into the affected tissue in RA and vasculitis and are also thought to contribute to the activation of leukocytes once in the tissue, resulting in integrin activation and the production of proteases and inflammatory mediators [\(5\)](#). Thus, the control of leukocyte entry into the tissue represents a major point at which new therapeutics, such as blockade of the chemokine system, could be developed to attenuate inflammation in RA and vasculitis.

In this review, we summarize the pathogenic roles of chemokines and their receptors in RA and vasculitis. Additionally, we provide an update on the clinical trials of chemokine- and chemokine-receptor-targeting drugs in arthritis and vasculitis and discuss their potential as therapeutic targets.

2. Rheumatoid Arthritis

Inflammatory arthritis, including rheumatoid arthritis (RA), is characterized by immune cell infiltration into joints, resulting in pannus formation with associated cartilage and bone destruction. This immune cell infiltrate drives RA pathogenesis and is composed of multiple cell types, including lymphocytes, macrophages, and neutrophils. Approximately 1 million people in Japan have RA. If not effectively treated, RA causes irreversible joint destruction resulting in significant disability, increased risk of cardiovascular disease, and markedly increased medical costs. Biological therapies, such as TNF inhibitors and IL-6 inhibitors, have revolutionized the treatment of RA. However, ~50% of RA patients still do not respond to these agents, and patients who do respond often have a residual disease and are at an increased risk of infection, such as pneumonia, associated with their use [\(6\)](#). Therefore, there remains an ongoing need to identify new therapeutic targets and treatment strategies for RA.

Animal models of inflammatory arthritis have provided valuable research tools for understanding the pathogenic mechanism of RA and for studying therapeutic targets. In fact, several animal models of inflammatory arthritis, such as the K/BxN arthritogenic serum transfer model of arthritis [\(7\)](#), the type II collagen-induced arthritis (CIA) [\(8\)](#) and the collagen antibody-induced arthritis (CAIA) [\(9\)](#), have a number of similarities to RA, including synovial hyperplasia, mononuclear cell infiltration and bone destruction. Therefore, these animal models have been used to define the role of chemokines and their receptors in inflammatory arthritis. In this section, we describe the functional role of chemokines and their receptors in animal models and in human RA and discuss therapies targeting these molecules.

2.1 Chemokine in RA

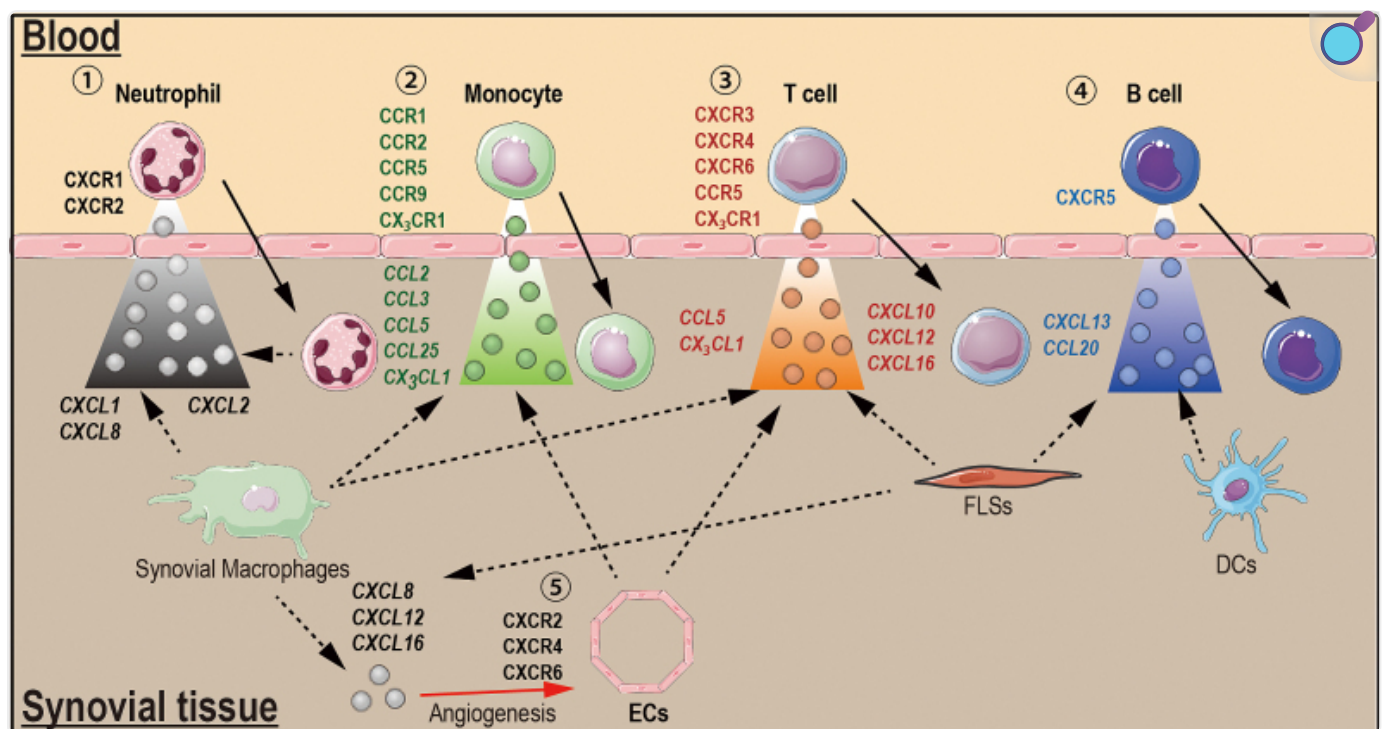
Numerous studies have demonstrated that multiple chemokines are highly expressed in serum, synovial fluid and synovial tissue of patients with RA, compared to healthy controls. For instance, the CXC-chemokines (CXCL5, CXCL8, CXCL9, CXCL10, CXCL12, CXCL13 and CXCL16), the CC-chemokines (CCL2, CCL3, CCL4, CCL5, CCL18, CCL19, CCL20, CCL21, CCL25), and the CX₃C-chemokine CX₃CL1 are increased in serum, synovial fluid, and synovial tissue of RA patients, compared with healthy controls [\(10\)](#), [\(11\)](#), [\(12\)](#), [\(13\)](#), [\(14\)](#), [\(15\)](#). In the pathogenesis of human RA, CXC-chemokines promote neutrophils (CXCL1, CXCL2, CXCL5 and CXCL8) [\(16\)](#), the effector T cell (CXCL10) [\(17\)](#) and the B cell (CXCL13) [\(18\)](#), [\(19\)](#) recruitment into the joint. CC-chemokines induce monocytes (CCL2, CCL3, CCL4, CCL5 and CCL7) [\(20\)](#), [\(21\)](#), T cells (CCL18, CCL19, CCL20, CCL21 and CCL25) [\(22\)](#) and the B cell (CCL20) [\(23\)](#) entry into the inflamed joint, whereas CX₃CL1 induces monocyte recruitment [\(24\)](#).

In addition, CXCL8, CXCL12 and CXCL16 induce angiogenesis in synovial tissue in RA, and CXCL8, CXCL10 and CXCL13 are promising biomarkers of RA disease activity/severity [\(11\)](#), [\(25\)](#). Synovial macrophages and fibroblast-like synoviocytes (FLSs) might be the main producers of

many inflammatory chemokines in the joints of RA patients, whereas synovial endothelial cells mainly produce CX₃CL1 and some CC-chemokines in RA patients. However, the pathogenic roles of chemokines in human RA remain unclear.

These same chemokines are also elevated in the serum, synovial fluid and synovial tissue in animal models of inflammatory arthritis, compared with control animals [\(20\)](#), [\(21\)](#), [\(22\)](#), [\(24\)](#), [\(26\)](#), [\(27\)](#), [\(28\)](#), [\(29\)](#), [\(30\)](#), [\(31\)](#), [\(32\)](#). Thus, animal models of arthritis are available to dissect the pathogenic roles of chemokines in inflammatory arthritis. Several studies using the animal models have indicated the development of arthritis in RA: (i) immune complexes stimulate synovial macrophages to generate CXCL1 that initiates neutrophil recruitment into the joints in the early phase of animal models of arthritis [\(33\)](#); (ii) after neutrophils enter the inflamed tissue, they become activated and produce CXCL2, which contributes to the recruitment of additional neutrophils in a positive-feedback loop [\(26\)](#), [\(33\)](#); (iii) neutrophils can also amplify the local response by secreting inflammatory cytokines, such as IL-1, that induce chemokine production by FLSs in the joint [\(26\)](#); (iv) In later phases of the disease, FLSs and other cells might play important roles in recruiting T and B cells ([Figure 1](#)).

Figure 1.


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Chemokines and chemokine receptors in rheumatoid arthritis.

① Neutrophil recruitment: Synovial macrophages generate CXCL1 and CXCL8, which promote neutrophil recruitment into the joint. Once neutrophils enter into the joint, they become activated and produce CXCL2, which amplifies neutrophil recruitment and stimulates additional chemokine production by macrophages. ② Monocyte recruitment: Synovial macrophages generate chemokines that promote monocyte recruitment through CCR2, CCR5, CCR9 and CX₃CR1. ③ T cell recruitment: Synovial macrophages, endothelial cells (ECs) and fibroblast-like synovial cells (FLSs) generate chemokines that induce T cell trafficking through CXCR3, CXCR4, CXCR6, CCR5 and CX₃CR1. ④ B cell recruitment; FLSs and dendritic cells (DCs) produce CXCL13 and CCL20 that promote B cell recruitment through CXCR5. ⑤ Angiogenesis: CXCL8, CXCL12 and CXCL16 produced by synovial macrophage and FLSs induce angiogenesis through CXCR2, CXCR4, and CXCR6, respectively, expressed on ECs. Solid arrows depict cell movement in response to chemokines, and dashed arrows indicate chemokine secretion from cell. The red solid arrow indicates chemokine-inducing angiogenesis.

2.2 Chemokine receptors in RA

Multiple chemokine receptors (CXCR1-6, CCR1, CCR2, CCR5-7, CCR9 and CCR10) and chemokines are all highly expressed in the joints of human RA and in animal models of arthritis [\(10\)](#), [\(20\)](#), [\(22\)](#), [\(26\)](#), [\(27\)](#), [\(34\)](#), [\(35\)](#). In addition, chemokine receptors control the recruitment of neutrophils (CXCR1 and CXCR2), T helper type 1 (T_H1) cell (CXCR3), lymphocyte (CXCR4), B cell (CXCR5), T_{FH} cells (CXCR5), T cell (CXCR6, CCR5 and CCR6), monocyte (CCR1, CCR2, CCR5-7, CCR9 and CCR10) and DCs (CCR9) into the inflamed joints in human RA and in animal models [\(Figure 1\)](#) [\(10\)](#), [\(17\)](#), [\(18\)](#), [\(19\)](#), [\(20\)](#), [\(22\)](#), [\(23\)](#), [\(26\)](#), [\(35\)](#), [\(36\)](#). The CX₃C-chemokine receptor CX₃CR1 is expressed on CD3-positive T cells in the RA synovium [\(15\)](#), suggesting that the receptor also contributes to the recruitment of T cells into RA joints [\(Figure 1\)](#). Thus, different combinations of chemokine receptors are required for the entry of different cell types into the synovium during RA. For instance, CXCR1 and CXCR2 are required for neutrophils, CCR2, CCR5, CCR9, and CX₃CR1 for monocytes; CXCR3, CXCR4, CXCR6, CCR5, CCR6 and CX₃CR1 for T cells; and CXCR5 for B cells [\(Figure 1\)](#).

Some chemokine receptors are also expressed on stromal cells in RA patients, although their functions on these cells are largely unknown. Although chemokine receptor signaling in synovial endothelial cells (ECs) during RA may be an important mediator of angiogenesis, others in RA FLSs may contribute to induce the production of inflammatory mediators. For instance, CCL21/CCR7 and CCL28/CCR10 signaling in synovial ECs promotes RA angiogenesis [\(37\)](#), [\(38\)](#). CCR9, which is highly expressed on FLSs during RA, induces the production of inflammatory cytokines *in vitro* upon stimulation with CCL25, although the role of this interaction *in vivo* is not yet clear [\(22\)](#). These studies suggest that in addition to immune cells, the chemokine system also plays important roles in the function of stromal cells, such as ECs and FLSs, in the pathogenesis of RA.

Atypical chemokine receptors (ACKRs) are homologous to *classical* chemokine receptors and bind chemokines but do not couple to G proteins. These atypical receptors appear to play a primary role in shaping chemokine gradients by scavenging and controlling transcytosis of chemokines. Atypical chemokine receptor 1 (ACKR1/DARC) and ACKR3/CXCR7 are expressed on joint ECs in human RA synovial tissue [\(13\)](#), [\(39\)](#), [\(40\)](#). It has recently been reported that ACKR2/D6 expression is elevated on peripheral blood cells and on leukocytes and stromal cells in synovial tissue of RA patients [\(41\)](#). More recently, we have demonstrated that ACKR1/DARC has transcytosed inflammatory chemokines (CXCL1 and CXCL2) to initiate neutrophil entry into the joint in animal models of inflammatory arthritis [\(39\)](#), [\(42\)](#). Meanwhile, ACKR3/CXCR7 has been demonstrated to play a role in angiogenesis in an animal model of arthritis [\(13\)](#). However, the functional role of ACKRs in human RA pathogenesis remains unknown.

2.3 Targeting the chemokine system in RA

Numerous chemokines and their receptor inhibitors have been investigated and tested in the animal models. For instance, blockade of a single chemokine (CXCL10, CXCL13, CCL2, CX₃CL1 or XCL1) or chemokine receptors (CXCR2, CXCR3, CXCR5, CXCR7, CCR1, CCR2, CCR5, CCR7 and CCR9) have demonstrated preventive and, in some cases, therapeutic efficacy in animal models [\(13\)](#), [\(17\)](#), [\(21\)](#), [\(22\)](#), [\(24\)](#), [\(26\)](#), [\(27\)](#), [\(36\)](#), [\(43\)](#), [\(44\)](#), [\(45\)](#), [\(46\)](#), [\(47\)](#), [\(48\)](#), [\(49\)](#), [\(50\)](#), [\(51\)](#), [\(52\)](#), [\(53\)](#), [\(54\)](#).

However, only a few clinical trials (CXCL10; MDX-10 [\(14\)](#), CX3CL1; E6011 and CCR1; CCX354-C [\(55\)](#)) of 10 drugs that target the chemokine system had some clinical benefits in RA patients ([Table 3](#)).

Table 3.

Targeting the Chemokine Systems in Rheumatoid Arthritis.

Target	Drug (Type of drug)	Type of study	Efficacy	Study outcome
CXCL10	MDX-1100 (antibody)	Phase II	Mildly effective	The ACR20 response at week 12 was 54% (MDX1100 and MTX) and 17% (placebo and MTX)
CCL2	ABN912 (antibody)	Phase Ib	Not effective	ABN912 did not result in any clinical improvement.
CCR1	CP-481,715 (small molecules)	Phase Ib	Mildly effective	CP-481,715 reduced tender and swollen joint count, and macrophages infiltration into the synovial tissue than those of placebo.
	CCX354-C (small molecules)	Phase II	Mildly effective	The ACR20 response at week 12 was 39% (placebo), 43% (CCX354-C; 100mg twice daily) and 52% (CCX354-C; 200 mg once daily)
	MLN3897 (small molecules)	Phase IIa	Not effective	The ACR20 response at week 12 was 35% (MLN3897) and 33% (placebo).
CCR2	MLN1202 (antibody)	Phase IIa	Not effective	Patients treated with CCR2 monoclonal antibody or placebo for 6 weeks. No clinical improvement
CCR5	SCH351125 (small molecules)	Phase Ib	Not effective	The ACR20 response at week 4 was 20% (SCH351125) and 33% (placebo).
	AZD5672 (small molecules)	Phase IIb	Not effective	The ACR20 response at week 12 was around 35% (AZD5672) and 38% (placebo).
	UK-427,857 (small molecules)	Phase IIa	Not effective	The ACR20 response at week 12 was 23.7% (UK-427,857) and 23.8% (placebo).

Target	Drug (Type of drug)	Type of study	Efficacy	Study outcome
CX ₃ CL1	E6011 (antibody)	Phase I/II	Effective? (no placebo)	~60% treated patients had at ACR20 response at week 12.

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This table is modified from Miyabe Y, Lian J, Miyabe C, et al. Chemokines in rheumatic diseases: pathogenic role and therapeutic implications. *Nat Rev Rheumatol*. 2019;15(12):731-46 [\(5\)](#). The authors have the right to use the original table in Reference 5 and got the permission from Springer Nature.

Despite many inhibitors targeting the chemokine system failed to show positive results in clinical trials, indicating that the chemokine system might not be therapeutic targets for RA may be premature. CCR1 inhibitors might be an instructive example. MLN3897 was found to have a much lower pharmacokinetic and pharmacodynamic profile for CCR1 blockade compared to CCX354-C [\(56\)](#), suggesting that the lack of success of MLN3897 was not necessarily due to CCR1 being a poor target in terms of its underlying biology. Thus, CP-481,715 and MLN3897 failed to show any beneficial effects in clinical trials [\(Table 3\)](#) [\(55\)](#). It will be also important for inhibitors targeting the chemokine system to determine if optimum pharmacokinetic properties and receptor inhibition have been achieved before making final conclusions regarding the usefulness of inhibiting that particular target in a given disease process.

In addition, a more recent study of the animal model demonstrated that broadly cross-reactive chemokine-blocking antibodies for CXCR2 ligands (CXCL1-3 and CXCL5) dramatically attenuated inflammatory arthritis compared with blocking antibodies against a single chemokine (CXCL1) [\(57\)](#). Because the functions, such as adhesion and extravasation, during leukocyte trafficking overlaps between many chemokine systems, inhibition of a single chemokine system alone might not be sufficient to completely suppress leukocyte recruitment. Moreover, the pathogenesis of human RA is undoubtedly more complex than animal models of arthritis. Thus, strategies that inhibit *multiple* inflammatory chemokines, rather than a *single* chemokine alone, might also be a more promising direction for new RA therapies.

3. Vasculitis

Vasculitis is an autoimmune disease characterized by the presence of inflammatory leukocytes in vessel walls associated with destructive damage to vessel structures. The affected vessels vary in size, type, and location according to the type of vasculitic disease; (i) small vessels, e.g., ANCA-associated vasculitis (AAV); (ii) medium vessels, e.g., Kawasaki disease (KD); and (iii) large vessels, e.g., Takayasu disease (TD) and giant cell arteritis (GCA) [\(58\)](#). Corticosteroids and immunosuppressive agents, such as cyclophosphamide, methotrexate and tacrolimus, are commonly used for the treatment of vasculitis. However, in some cases, the disease is refractory to these treatments, and immunosuppression often leads to significant clinical complications. Therefore, there is a need for new therapies for vasculitis that are safer and more effective than currently used treatments.

Several experimental models of vasculitis have been developed, including *Polyoma* virus infection-induced vasculitis, ANCA-induced vasculitis and *Lactobacillus casei*-induced vasculitis, but the ability to induce vasculitis in these models is limited depending on the genetic background of the mouse strain [\(59\)](#). Injection of a *Candida albicans* water-soluble fraction (CAWS) into mice leads to inflammation of the aortic root and coronary arteries and can be used as a model for KD. CAWS-induced vasculitis can be induced in a variety of genetic backgrounds and is, therefore, more versatile compared to other animal models of studying the pathogenesis of vasculitis [\(59\)](#), [\(60\)](#). In CAWS-induced vasculitis, multiple immune cells such as neutrophils, macrophages and T cells, infiltrated the inflamed sites. Additionally, neutrophils might be a key driver of inflammation in CAWS-induced vasculitis [\(60\)](#).

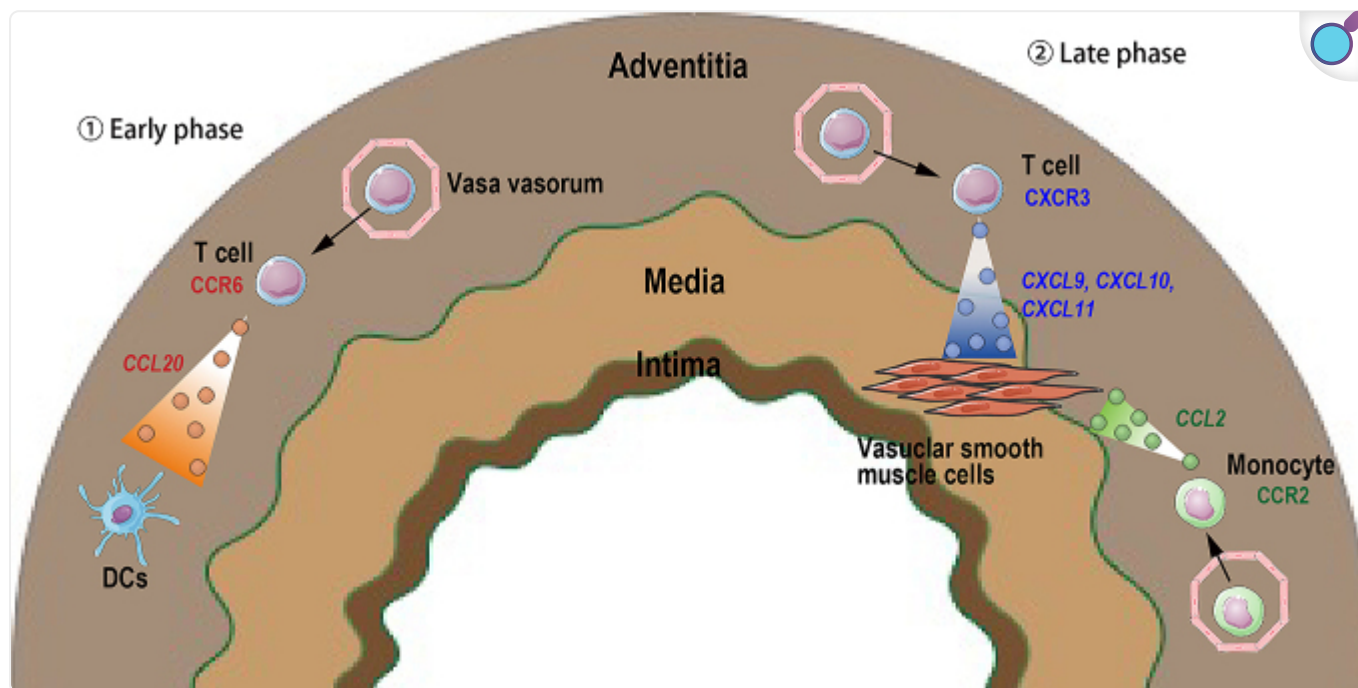
3.1 Chemokines in vasculitis

Several chemokines are elevated in the plasma and serum of vasculitis patients and are correlated with disease activity (AAV: CCL17 [\(61\)](#), CCL18 [\(62\)](#), CCL20 [\(63\)](#), CXCL8-11 [\(4\)](#), CX₃CL1 [\(64\)](#), and XCL1 [\(65\)](#); TD: CCL2 [\(66\)](#) and CCL5 [\(67\)](#); GCA: CCL2 [\(67\)](#), CXCL9-11 [\(68\)](#) and CX₃CL1 [\(69\)](#); and KD: CCL17 [\(70\)](#) and CXCL9/10 [\(70\)](#), [\(71\)](#)). In addition, CCL2, CCL7, CXCL2-3 and CXCL9-10 are highly expressed in the aortic root and coronary arteries of the CAWS-induced vasculitis mouse model [\(72\)](#), [\(73\)](#). Thus, the chemokine system also contributes to the development of vasculitis and RA.

In GCA, tissue-resident DCs in the adventitia of affected arteries may be a major chemokine producer in the early phase of disease onset, whereas vascular smooth muscle cells and inflammatory monocytes recruited into the artery generate chemokines during later phases [\(Figure 2\)](#) [\(74\)](#). More recently, we have demonstrated that Dectin-2-mediated CCL2 produced by tissue-resident macrophages in the aortic root and coronary arteries may be the key initiator of vascular inflammation in the CAWS-induced vasculitis. Thus, tissue-resident immune cells, such as tissue-resident macrophages and DCs, might be the main source of chemokines igniting

vascular inflammation in the development of GCA and KD. However, the main sources and pathogenic role of chemokine system in other forms of human vasculitis and animal models are still unknown.

Figure 2.



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Chemokines and chemokine receptors in giant cell arteritis.

① Early phase: Tissue-resident dendritic cells (DCs) in adventitia produce CCL20 that promotes CCR6⁺ T cell infiltration. ② Late phase: Vascular smooth muscle cells in media produce CXCL9, CXCL10 and CXCL11, which attract CXCR3⁺ T cells. Vascular smooth muscle cells also produce CCL2 resulting in CCR2⁺ monocyte recruitment. Solid arrows depict cell movement in response to chemokines, and dashed arrows indicate chemokine secretion from a cell.

3.2 Chemokine receptors in vasculitis

CCR2 and CX₃CR1 expression on monocytes and CXCR3 and CCR6 expression on CD8⁺ T cells, along with high levels of the ligands for these receptors, can be observed in temporal arterial biopsies derived from patients with active GCA ([Figure 2](#)) ^{(68), (69)}. In addition, expression of

CXCR2 and CCR2 in the aortic root and coronary arteries is increased in CAWS-induced vasculitis mice, suggesting that CXCR2 promotes neutrophil recruitment and CCR2 controls monocyte infiltration. However, the patterns of chemokine receptor expression and their pathogenic role in human and animal models of vasculitis have not been well studied yet.

3.3 Targeting the chemokine systems in vasculitis

CCR2-deficient mice are resistant to the development of CAWS-induced vasculitis, suggesting that chemokines and their receptors might be attractive new targets for vasculitis treatments [\(75\)](#). However, the effect of inhibiting chemokines and chemokine receptors in human vasculitis and in animal models has not been well studied yet. Recently, a complementary component C5a that is a chemoattractant, but not a chemokine, can be a new promising target for vasculitis. In fact, the C5a receptor inhibitor looks very promising in the clinical trial of ANCA-associated vasculitis [\(76\)](#). Thus, additional studies are required to establish a new vasculitis therapy for targeting the chemokine system.

4. Conclusion

Many chemokines and their receptors have been implicated in leukocyte recruitment into inflamed tissue in RA and vasculitis. Thus, chemokines and their receptors represent promising targets for therapeutic intervention in the treatment of RA and vasculitis. In fact, inhibition of chemokines and their receptors ameliorates inflammation in animal models. However, clinical trials with chemokine and chemokine receptor blockers in human RA have failed to show clinical effects. In addition, the mechanism of action for many chemokines and their receptors in RA and vasculitis is still unknown. An important observation that requires further understanding is that different chemokines may be important at different stages of disease depending on RA and vasculitis. Recent advances in imaging technology have provided unprecedented views into immune cell function in live animals, providing an entirely new paradigm for immune cell function. In fact, our recent work using multiphoton intravital microscopy to study neutrophil entry into the joint in the KxB/N model demonstrated that the *classical* C5a receptor C5aR1 and the *atypical* C5a receptor C5aR2 were required for initial integrin-dependent neutrophil arrest on the joint endothelium but were not involved in inducing neutrophil extravasation [\(16\)](#), [\(42\)](#). By contrast, the classical chemokine receptor CXCR2 and the atypical chemokine receptor ACKR1 were involved in neutrophil diapedesis into the joint space [\(16\)](#), [\(42\)](#). In addition, other work has indicated that blockade of multiple chemokines might be more effective than single chemokine inhibition, because functions of some chemokines in immune cell trafficking overlap [\(57\)](#). These studies point to a need to more fully dissect the functional roles of chemokines and their receptors in the pathogenesis of RA and vasculitis. A significant need remains for the treatment of RA and vasculitis. Recent clinical

studies suggest that the development of more effective inhibitors of chemokines and their receptors has untapped therapeutic potential.

Article Information

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Conflicts of Interest

None

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References

1. Nourshargh S, Alon R. Leukocyte migration into inflamed tissues. *Immunity*. 2014;41(5):694-707. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
2. Griffith JW, Sokol CL, Luster AD. Chemokines and chemokine receptors: positioning cells for host defense and immunity. *Annu Rev Immunol*. 2014;32:659-702. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Szekanecz Z, Koch AE. Successes and failures of chemokine-pathway targeting in rheumatoid arthritis. *Nat Rev Rheumatol*. 2016;12(1):5-13. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
4. Berti A, Warner R, Johnson K, et al. Brief report: circulating cytokine profiles and antineutrophil cytoplasmic antibody specificity in patients with antineutrophil cytoplasmic antibody-associated vasculitis. *Arthritis Rheumatol*. 2018;70(7):1114-21. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

5. Miyabe Y, Lian J, Miyabe C, et al. Chemokines in rheumatic diseases: pathogenic role and therapeutic implications. *Nat Rev Rheumatol*. 2019;15(12):731-46. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Komano Y, Tanaka M, Nanki T, et al. Incidence and risk factors for serious infection in patients with rheumatoid arthritis treated with tumor necrosis factor inhibitors: a report from the Registry of Japanese Rheumatoid Arthritis Patients for Longterm Safety. *J Rheumatol*. 2011;38(7). [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
7. Monach PA, Mathis D, Benoist C. The K/BxN arthritis model. *Curr Protoc Immunol*. 2008;Chapter(15):Unit 15.22. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Brand DD, Latham KA, Rosloniec EF. Collagen-induced arthritis. *Nat Protoc*. 2007;2(5):1269-75. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Khachigian LM. Collagen antibody-induced arthritis. *Nat Protoc*. 2006;1(5):2512-6. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Nanki T, Shimaoka T, Hayashida K, et al. Pathogenic role of the CXCL16-CXCR6 pathway in rheumatoid arthritis. *Arthritis Rheum*. 2005;52(10):3004-14. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
11. Meeuwisse CM, van der Linden MP, Rullmann TA, et al. Identification of CXCL13 as a marker for rheumatoid arthritis outcome using an in silico model of the rheumatic joint. *Arthritis Rheum*. 2011;63(5):1265-73. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
12. Endo H, Akahoshi T, Takagishi K, et al. Elevation of interleukin-8 (IL-8) levels in joint fluids of patients with rheumatoid arthritis and the induction by IL-8 of leukocyte infiltration and synovitis in rabbit joints. *Lymphokine Cytokine Res*. 1991;10(4):245-52. [[PubMed](#)] [[Google Scholar](#)]
13. Watanabe K, Penfold ME, Matsuda A, et al. Pathogenic role of CXCR7 in rheumatoid arthritis. *Arthritis Rheum*. 2010;62(11):3211-20. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
14. Yellin M, Paliienko I, Balanescu A, et al. A phase II, randomized, double-blind, placebo-controlled study evaluating the efficacy and safety of MDX-1100, a fully human anti-CXCL10 monoclonal antibody, in combination with methotrexate in patients with rheumatoid arthritis. *Arthritis Rheum*. 2012;64(6):1730-9. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
15. Nanki T, Imai T, Nagasaka K, et al. Migration of CX3CR1-positive T cells producing type 1 cytokines and cytotoxic molecules into the synovium of patients with rheumatoid arthritis. *Arthritis Rheum*. 2002;46(11):2878-83. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

16. Miyabe Y, Miyabe C, Murooka TT, et al. Complement C5a receptor is the key initiator of neutrophil adhesion igniting immune complex-induced arthritis. *Sci Immunol*. 2017;2(7). [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
17. Lee JH, Kim B, Jin WJ, et al. Pathogenic roles of CXCL10 signaling through CXCR3 and TLR4 in macrophages and T cells: relevance for arthritis. *Arthritis Res Ther*. 2017;19(1):163. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
18. Armas-González E, Domínguez-Luis MJ, Díaz-Martín A, et al. Role of CXCL13 and CCL20 in the recruitment of B cells to inflammatory foci in chronic arthritis. *Arthritis Res Ther*. 2018;20(1):114. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
19. Moschovakis GL, Bubke A, Friedrichsen M, et al. T cell specific Cxcr5 deficiency prevents rheumatoid arthritis. *Sci Rep*. 2017;7(1):8933. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
20. Haringman JJ, Ludikhuize J, Tak PP. Chemokines in joint disease: the key to inflammation? *Ann Rheum Dis*. 2004;63(10):1186-94. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
21. Barnes DA, Tse J, Kaufhold M, et al. Polyclonal antibody directed against human RANTES ameliorates disease in the Lewis rat adjuvant-induced arthritis model. *J Clin Invest*. 1998;101(12):2910-9. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
22. Yokoyama W, Kohsaka H, Kaneko K, et al. Abrogation of CC chemokine receptor 9 ameliorates collagen-induced arthritis of mice. *Arthritis Res Ther*. 2014;16(5):445. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
23. Nanki T, Takada K, Komano Y, et al. Chemokine receptor expression and functional effects of chemokines on B cells: implication in the pathogenesis of rheumatoid arthritis. *Arthritis Res Ther*. 2009;11(5):R149. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
24. Nanki T, Urasaki Y, Imai T, et al. Inhibition of fractalkine ameliorates murine collagen-induced arthritis. *J Immunol*. 2004;173(11):7010-6. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
25. Pandya JM, Lundell AC, Andersson K, et al. Blood chemokine profile in untreated early rheumatoid arthritis: CXCL10 as a disease activity marker. *Arthritis Res Ther*. 2017;19(1):20. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
26. Chou RC, Kim ND, Sadik CD, et al. Lipid-cytokine-chemokine cascade drives neutrophil recruitment in a murine model of inflammatory arthritis. *Immunity*.

2010;33(2):266-78. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

27. Isozaki T, Arbab AS, Haas CS, et al. Evidence that CXCL16 is a potent mediator of angiogenesis and is involved in endothelial progenitor cell chemotaxis: studies in mice with K/BxN serum-induced arthritis. *Arthritis Rheum.* 2013;65(7):1736-46. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

28. Zheng B, Ozen Z, Zhang X, et al. CXCL13 neutralization reduces the severity of collagen-induced arthritis. *Arthritis Rheum.* 2005;52(2):620-6. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

29. Jacobs JP, Ortiz-Lopez A, Campbell JJ, et al. Deficiency of CXCR2, but not other chemokine receptors, attenuates autoantibody-mediated arthritis in a murine model. *Arthritis Rheum.* 2010;62(7):1921-32. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

30. García-Vicuña R, Gómez-Gaviro MV, Domínguez-Luis MJ, et al. CC and CXC chemokine receptors mediate migration, proliferation, and matrix metalloproteinase production by fibroblast-like synoviocytes from rheumatoid arthritis patients. *Arthritis Rheum.* 2004;50(12):3866-77. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

31. Takayasu A, Miyabe Y, Yokoyama W, et al. CCL18 activates fibroblast-like synoviocytes in patients with rheumatoid arthritis. *J Rheumatol.* 2013;40(6):1026-8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

32. Pickens SR, Chamberlain ND, Volin MV, et al. Characterization of CCL19 and CCL21 in rheumatoid arthritis. *Arthritis Rheum.* 2011;63(4):914-22. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

33. Li JL, Lim CH, Tay FW, et al. Neutrophils self-regulate immune complex-mediated cutaneous inflammation through CXCL2. *J Invest Dermatol.* 2016;136(2):416-24. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

34. Patterson AM, Schmutz C, Davis S, et al. Differential binding of chemokines to macrophages and neutrophils in the human inflamed synovium. *Arthritis Res.* 2002;4(3):209-14. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

35. Ruth JH, Rottman JB, Katschke KJ, et al. Selective lymphocyte chemokine receptor expression in the rheumatoid joint. *Arthritis Rheum.* 2001;44(12):2750-60. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

36. Wengner AM, Höpken UE, Petrow PK, et al. CXCR5- and CCR7-dependent lymphoid neogenesis in a murine model of chronic antigen-induced arthritis. *Arthritis Rheum.*

2007;56(10):3271-83. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

37. Pickens SR, Chamberlain ND, Volin MV, et al. Role of the CCL21 and CCR7 pathways in rheumatoid arthritis angiogenesis. *Arthritis Rheum.* 2012;64(8):2471-81. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

38. Chen Z, Kim SJ, Essani AB, et al. Characterising the expression and function of CCL28 and its corresponding receptor, CCR10, in RA pathogenesis. *Ann Rheum Dis.* 2015;74(10):1898-906. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

39. Smith E, McGettrick HM, Stone MA, et al. Duffy antigen receptor for chemokines and CXCL5 are essential for the recruitment of neutrophils in a multicellular model of rheumatoid arthritis synovium. *Arthritis Rheum.* 2008;58(7):1968-73. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

40. Patterson AM, Siddall H, Chamberlain G, et al. Expression of the Duffy antigen/receptor for chemokines (DARC) by the inflamed synovial endothelium. *J Pathol.* 2002;197(1):108-16. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

41. Baldwin HM, Singh MD, Codullo V, et al. Elevated ACKR2 expression is a common feature of inflammatory arthropathies. *Rheumatol Oxf Engl.* 2017;56(9):1607-17. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

42. Miyabe Y, Miyabe C, Mani V, et al. Atypical complement receptor C5aR2 transports C5a to initiate neutrophil adhesion and inflammation. *Sci Immunol.* 2019;4(35). [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

43. Kim B, Lee JH, Jin WJ, et al. JN-2, a C-X-C motif chemokine receptor 3 antagonist, ameliorates arthritis progression in an animal model. *Eur J Pharmacol.* 2018;823:1-10. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

44. Talbot J, Bianchini FJ, Nascimento DC, et al. CCR2 expression in neutrophils plays a critical role in their migration into the joints in rheumatoid arthritis. *Arthritis Rheumatol.* 2015;67(7):1751-9. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

45. Amat M, Benjamim CF, Williams LM, et al. Pharmacological blockade of CCR1 ameliorates murine arthritis and alters cytokine networks in vivo. *Br J Pharmacol.* 2006;149(6):666-75. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

46. Brodmerkel CM, Huber R, Covington M, et al. Discovery and pharmacological characterization of a novel rodent-active CCR2 antagonist, INCB3344. *J Immunol.* 2005;175(8):5370-8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

47. Vierboom MP, Zavodny PJ, Chou CC, et al. Inhibition of the development of collagen-induced arthritis in rhesus monkeys by a small molecular weight antagonist of CCR5. *Arthritis Rheum.* 2005;52(2):627-36. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
48. Bonelli M, Puchner A, Göschl L, et al. CCR6 controls autoimmune but not innate immunity-driven experimental arthritis. *J Cell Mol Med.* 2018;22(11):5278-85. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
49. Moschovakis GL, Bubke A, Friedrichsen M, et al. The chemokine receptor CCR7 is a promising target for rheumatoid arthritis therapy. *Cell Mol Immunol.* 2019;16(10):791-9. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
50. Min SH, Wang Y, Gonsiorek W, et al. Pharmacological targeting reveals distinct roles for CXCR2/CXCR1 and CCR2 in a mouse model of arthritis. *Biochem Biophys Res Commun.* 2010;391(1):1080-6. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
51. Slauenwhite D, Gebremeskel S, Doucette CD, et al. Regulation of cytokine polarization and T cell recruitment to inflamed paws in mouse collagen-induced arthritis by the chemokine receptor CXCR6. *Arthritis Rheumatol.* 2014;66(11):3001-12. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
52. Matsumoto N, Kon S, Nakatsuru T, et al. A novel alpha 9 integrin ligand, XCL1/lymphotactin, is involved in the development of murine models of autoimmune diseases. *J Immunol.* 2017;199(1):82-90. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
53. Klimatcheva E, Pandina T, Reilly C, et al. CXCL13 antibody for the treatment of autoimmune disorders. *BMC Immunol.* 2015;16:6. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
54. Ogata H, Takeya M, Yoshimura T, et al. The role of monocyte chemoattractant protein-1 (MCP-1) in the pathogenesis of collagen-induced arthritis in rats. *J Pathol.* 1997;182(1):106-14. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
55. Tak PP, Balanescu A, Tseluyko V, et al. Chemokine receptor CCR1 antagonist CCX354-C treatment for rheumatoid arthritis: CARAT-2, a randomised, placebo controlled clinical trial. *Ann Rheum Dis.* 2013;72(3):337-44. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
56. Dairaghi DJ, Zhang P, Wang Y, et al. Pharmacokinetic and pharmacodynamic evaluation of the novel CCR1 antagonist CCX354 in healthy human subjects: implications for selection of clinical dose. *Clin Pharmacol Ther.* 2011;89(5):726-34. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

57. Angelini A, Miyabe Y, Newsted D, et al. Directed evolution of broadly crossreactive chemokine-blocking antibodies efficacious in arthritis. *Nat Commun*. 2018;9(1):1461. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
58. Watts RA, Scott DG. Recent developments in the classification and assessment of vasculitis. *Best Pract Res Clin Rheumatol*. 2009;23(3):429-43. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
59. Mogi M, Liu S. Animal models of vasculitis. *Methods Mol Biol*. 2018;1868:223-32. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
60. Miyabe C, Miyabe Y, Miura NN, et al. Am80, a retinoic acid receptor agonist, ameliorates murine vasculitis through the suppression of neutrophil migration and activation. *Arthritis Rheum*. 2013;65(2):503-12. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
61. Dallos T, Heiland GR, Strehl J, et al. CCL17/thymus and activation-related chemokine in Churg-Strauss syndrome. *Arthritis Rheum*. 2010;62(11):3496-503. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
62. Brix SR, Stege G, Disteldorf E, et al. CC chemokine ligand 18 in ANCA-associated crescentic GN. *J Am Soc Nephrol*. 2015;26(9):2105-17. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
63. Eriksson P, Andersson C, Cassel P, et al. Increase in Th17-associated CCL20 and decrease in Th2-associated CCL22 plasma chemokines in active ANCA-associated vasculitis. *Scand J Rheumatol*. 2015;44(1):80-3. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
64. Matsunawa M, Odai T, Wakabayashi K, et al. Elevated serum levels of soluble CX3CL1 in patients with microscopic polyangiitis. *Clin Exp Rheumatol*. 2009;27(1):72-8. [[PubMed](#)] [[Google Scholar](#)]
65. Blaschke S, Brandt P, Wessels JT, et al. Expression and function of the C-class chemokine lymphotactin (XCL1) in Wegener's granulomatosis. *J Rheumatol*. 2009;36(11):2491-500. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
66. Savioli B, Abdulahad WH, Brouwer E, et al. Are cytokines and chemokines suitable biomarkers for Takayasu arteritis? *Autoimmun Rev*. 2017;16(10):1071-8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
67. Dhawan V, Mahajan N, Jain S. Role of C-C chemokines in Takayasu's arteritis disease. *Int J Cardiol*. 2006;112(1):105-11. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
68. Samson M, Ly KH, Tournier B, et al. Involvement and prognosis value of CD8(+) T cells in giant cell arteritis. *J Autoimmun*. 2016;72:73-83. [[DOI](#)] [[PubMed](#)] [[Google](#)]

[Scholar](#)]

69. van Sleen Y, Wang Q, van der Geest KSM, et al. Involvement of monocyte subsets in the immunopathology of giant cell arteritis. *Sci Rep*. 2017;7(1):6553. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

70. Feng S, Yadav SK, Gao F, et al. Plasma levels of monokine induced by interferon-gamma/chemokine (C-X-X motif) ligand 9, thymus and activation-regulated chemokine/chemokine (C-C motif) ligand 17 in children with Kawasaki disease. *BMC Pediatr*. 2015;15:109. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

71. Ko TM, Kuo HC, Chang JS, et al. CXCL10/IP-10 is a biomarker and mediator for Kawasaki disease. *Circ Res*. 2015;116(5):876-83. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

72. Stock AT, Hansen JA, Sleeman MA, et al. GM-CSF primes cardiac inflammation in a mouse model of Kawasaki disease. *J Exp Med*. 2016;213(10):1983-98. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

73. Suzuki C, Nakamura A, Miura N, et al. Non-receptor type, proline-rich protein tyrosine kinase 2 (Pyk2) is a possible therapeutic target for Kawasaki disease. *Clin Immunol*. 2017;179:17-24. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

74. Samson M, Corbera-Bellalta M, Audia S, et al. Recent advances in our understanding of giant cell arteritis pathogenesis. *Autoimmun Rev*. 2017;16(8):833-44. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

75. Martinez HG, Quinones MP, Jimenez F, et al. Important role of CCR2 in a murine model of coronary vasculitis. *BMC Immunol*. 2012;13:56. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

76. Jayne DRW, Bruchfeld AN, Harper L, et al. Randomized trial of C5a receptor inhibitor Avacopan in ANCA-associated vasculitis. *J Am Soc Nephrol*. 2017;28(9):2756-67. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

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